Multiple IV Bolus Practice 4

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

A 192 lb patient is to receive IV bolus injections of a medication every 4 hours. The elimination half-life of the medication is approximately 3.3 hours and the apparent volume of distribution is 30% of body weight.

a Recommend a dose to achieve an average steady state concentration of 25 mg/L. (Round to the nearest 10 mg).

b What is the maximum plasma concentration of the first dose? (The dose is the 550 mg recommended in part a.)

c What is the maximum steady-state plasma concentration? (The dose is the 550 mg recommended in part a.)

d What is the minimum steady-state plasma concentration? (The dose is the 550 mg recommended in part a.)

e What is the plasma concentration 10 hours after the last dose assuming steady state was attained? (The dose is the 550 mg recommended in part a; part c: C_{max}^{∞} = 36.95 mg/L.)